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Descriptive Statistics & Inferential Statistics: The Complete Beginner-to-Intermediate Guide



Marishetty Ramyakrishna 19 min read · 21 hours ago



From understanding data to making reliable decisions from samples

What Is Statistics?

Statistics is the science of **collecting, organizing, analyzing, interpreting, and communicating data.**

In Data Science and Data Analytics, statistics helps us answer questions such as:

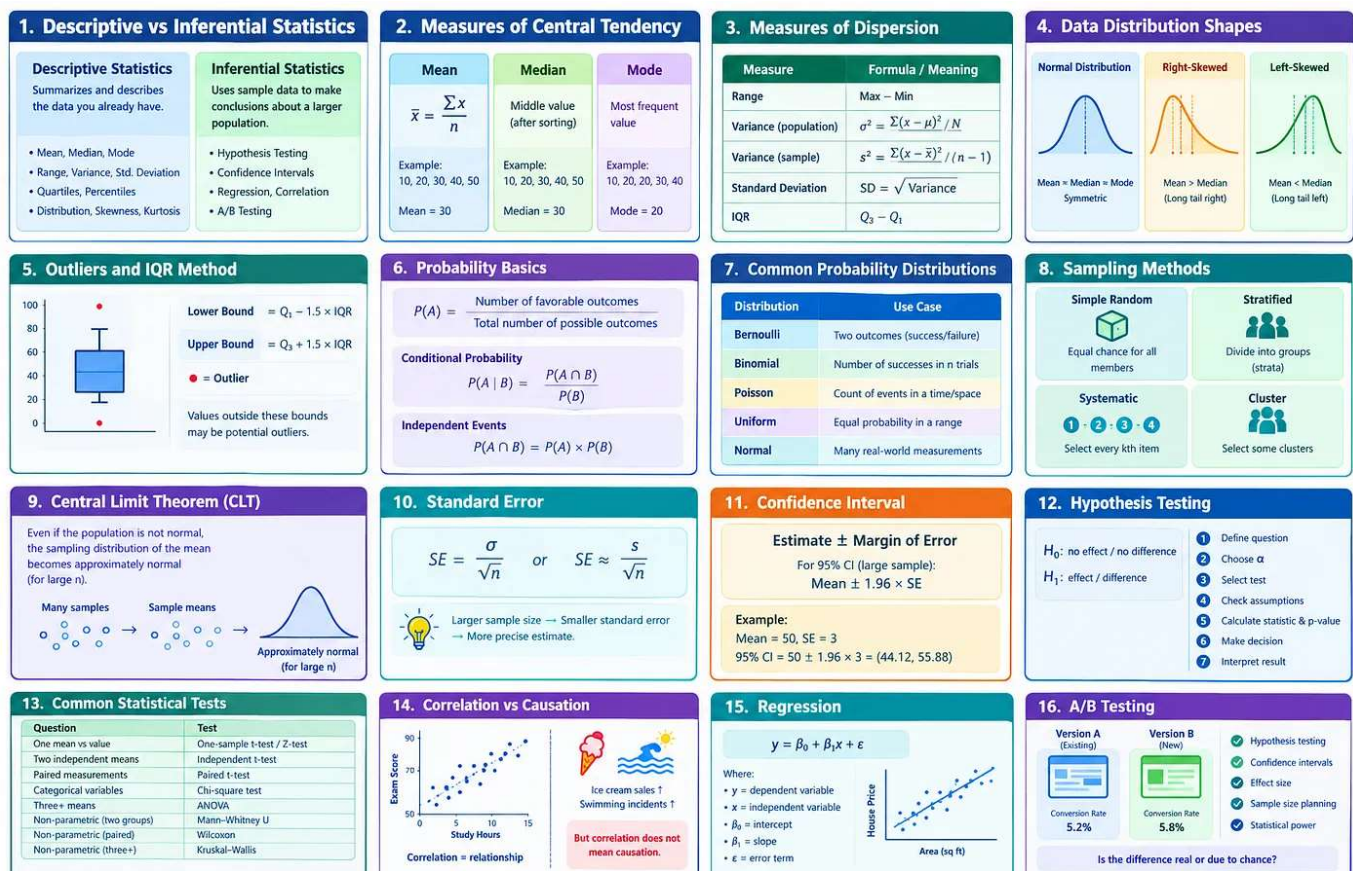
- What is the average value?
- How spread out is the data?
- Are there unusual values?

- What does the distribution look like?
- Can we make conclusions about a large population using a sample?
- Is an observed difference statistically significant?
- How confident are we about our estimate?
- Does one variable have a relationship with another?
- Can we use sample data to make business decisions?

Statistics is broadly divided into two major areas:

1. Descriptive Statistics

2. Inferential Statistics



1. Descriptive Statistics

Descriptive statistics is used to **summarize and describe the data we already have**.

It answers:

“What does my data look like?”

For example, suppose a company has the following customer ages:

20, 22, 25, 27, 30, 32, 35

We can calculate:

- Mean
- Median
- Minimum
- Maximum
- Range
- Variance
- Standard deviation
- Quartiles
- Percentiles

These values help us understand the dataset.

Important Point

Descriptive statistics **does not generalize beyond the data being analyzed.**

It describes the observed data.

Why Do We Need Descriptive Statistics?

Imagine you have a dataset containing **100,000 customer records.**

Looking at every row individually is difficult.

Instead, descriptive statistics gives us a quick summary.

For example:

Statistic	Value
Count	100,000
Mean Age	32.5
Median Age	31
Minimum Age	18
Maximum Age	75
Standard Deviation	8.2

Summary of Descriptive Statistics for the Age Variable

Now we immediately understand the basic characteristics of the dataset.

Population and Sample

Before learning inferential statistics, we need to understand two important terms.

Population

A **population** is the complete group we are interested in.

Example:

All customers of a bank.

Sample

A **sample** is a smaller group selected from the population.

Example:

1,000 customers selected from all customers of the bank.

Example:

Suppose a university has 50,000 students.

- Population = 50,000 students
- Sample = 1,000 students selected for a survey

We usually study a sample because studying the entire population can be expensive, slow, or impossible.

Parameter vs Statistic

These two terms are important in statistics.

Parameter

A **parameter** describes a population.

Examples:

- Population mean = μ
- Population standard deviation = σ

- Population proportion = p

Statistic

A **statistic** describes a sample.

Examples:

- Sample mean = $\bar{x}$
- Sample standard deviation = s
- Sample proportion = $\hat{p}$

Measures of Central Tendency

Central tendency tells us about the **center or typical value** of the data.

The three most common measures are:

1. Mean
2. Median
3. Mode

Mean

The mean is commonly called the **average**.

Formula

$$\text{Mean} = \Sigma x / n$$

Where:

- Σx = sum of all values
- n = number of observations

Example

Consider:

10, 20, 30, 40, 50

Sum:

$$10 + 20 + 30 + 40 + 50 = 150$$

Number of values:

5

Therefore:

$$\begin{aligned}\text{Mean} &= 150 / 5 \\ &= 30\end{aligned}$$

So the mean is 30.

Mean and Outliers

The mean is sensitive to extreme values.

Example:

10, 20, 30, 40, 500

Mean:

$$\begin{aligned}(10 + 20 + 30 + 40 + 500) / 5 \\ = 600 / 5 \\ = 120\end{aligned}$$

Although most values are between 10 and 40, the mean becomes 120 because of the extreme value 500.

This is why we should not blindly use the mean when the data contains strong outliers.

Median

The median is the **middle value** after sorting the data.

Example:

10, 20, 30, 40, 50

The middle value is:

30

Therefore:

Median = 30

Even Number of Values

Example:

10, 20, 30, 40

There are two middle values:

20 and 30

Median:

$$(20 + 30) / 2 = 25$$

Mean vs Median

Consider:

10, 20, 30, 40, 500

Mean:

120

Median:

30

he median better represents the typical value because it is less affected by the outlier.

Practical Rule

- Symmetric data → Mean can be useful
- Skewed data → Median is often more representative
- Data with strong outliers → Median can be more robust

Mode

The mode is the value that occurs **most frequently**.

Example:

10, 20, 20, 30, 40

20 occurs twice.

Therefore:

Mode = 20

Mode is especially useful for categorical data.

Example:

Red, Blue, Blue, Green, Blue

Mode:

Blue

Mean vs Median vs Mode

Measure	Meaning	Sensitive to Outliers?	Common Use
Mean	Average	Yes	Numerical data
Median	Middle value	No/less sensitive	Skewed data
Mode	Most frequent value	Generally no	Categorical/discrete data

difference between Mean, Median and Mode

Measures of Dispersion

Central tendency tells us where the data is centered.

But we also need to know:

How spread out is the data?

This is called **dispersion** or **variability**.

Important measures include:

- Range
- Variance
- Standard deviation
- Quartiles
- Percentiles
- Interquartile range

Range

Range is:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example:

10, 20, 30, 40, 50

Range:

$$50 - 10 = 40$$

Variance

Variance measures how far observations are spread around the mean.

Population Variance

$$\sigma^2 = \sum (x - \mu)^2 / N$$

Sample Variance

$$s^2 = \sum (x - \bar{x})^2 / (n - 1)$$

The difference between N and $n - 1$ is important when calculating variance from a sample.

Standard Deviation

Standard deviation is the square root of variance.

$$SD = \sqrt{\text{Variance}}$$

A smaller standard deviation means values are generally closer to the mean.

A larger standard deviation means values are more spread out.

Example

Dataset A:

48, 49, 50, 51, 52

Dataset B:

10, 30, 50, 70, 90

Both have a mean of 50, but Dataset B has much greater variability.

Quartiles

Quartiles divide sorted data into four parts.

- Q1 → 25th percentile

- $Q2 \rightarrow 50\text{th percentile}$
- $Q3 \rightarrow 75\text{th percentile}$

$Q2$ is the median.

Percentiles

A percentile tells us the position of a value relative to other observations.

For example:

A student at the 90th percentile performed better than approximately 90% of the reference group.

Percentiles are commonly used in:

- Entrance exams
- Salary analysis
- Customer analytics
- Performance analysis
- Healthcare
- Machine Learning

Interquartile Range (IQR)

IQR measures the spread of the middle 50% of the data.

Formula

$$\text{IQR} = Q3 - Q1$$

It is less affected by extreme values than the range.

Outliers

An outlier is an observation that is unusually far from the rest of the data.

The common IQR method uses:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Values outside these boundaries are often flagged as potential outliers.

Important

An outlier should **not automatically be deleted**.

It could represent:

- A data-entry error
- A genuine rare event
- Fraud
- A high-value customer

- A medical emergency
- An unusual business transaction

Always investigate the reason first.

Understanding Data Distribution

A distribution tells us how values are arranged.

Some common shapes include:

- Normal distribution
- Right-skewed distribution
- Left-skewed distribution

Normal Distribution

A normal distribution is approximately bell-shaped and symmetric.

Characteristics:

- Mean $\approx$ Median $\approx$ Mode
- Symmetric around the center
- Many observations occur around the mean
- Fewer observations occur toward the extremes

The standard normal distribution has:

Mean = 0
Standard Deviation = 1

Right-Skewed Distribution

In a right-skewed distribution, the long tail extends toward larger values.

Typically:

Mean > Median

Examples can include:

- Income
- House prices
- Certain transaction amounts

Left-Skewed Distribution

In a left-skewed distribution, the long tail extends toward smaller values.

Typically:

Mean < Median

Skewness

Skewness describes the asymmetry of a distribution.

- Skewness $\approx 0 \rightarrow$ approximately symmetric
- Positive skewness $\rightarrow$ right-skewed
- Negative skewness $\rightarrow$ left-skewed

Kurtosis

Kurtosis describes aspects of the **tail behavior and extremity** of a distribution.

It can help identify whether a distribution has relatively heavy or light tails compared with a reference distribution.

In practical data analysis, kurtosis can be useful when investigating unusual extreme observations.

Descriptive Statistics Using Pandas

Python's Pandas library makes descriptive statistics very easy.

```
import pandas as pd
```

```
df = pd.read_csv("data.csv")  
print(df.describe())
```

For a particular column:

```
print(df["Age"].mean())  
print(df["Age"].median())  
print(df["Age"].mode())  
print(df["Age"].std())  
print(df["Age"].var())  
print(df["Age"].min())  
print(df["Age"].max())
```

You can also use:

```
print(df["Age"].quantile(0.25))  
print(df["Age"].quantile(0.50))  
print(df["Age"].quantile(0.75))
```

Probability: The Foundation of Inferential Statistics

Probability measures the likelihood of an event.

The basic formula is:

$$P(A) = \text{Number of favorable outcomes} / \text{Total number of possible outcomes}$$

Probability ranges from:

0 to 1

Where:

- 0 → impossible
- 1 → certain

Conditional Probability

Conditional probability asks:

What is the probability of A happening given that B has already happened?

Formula

$$P(A \mid B) = P(A \cap B) / P(B)$$

Example:

Suppose we want to know the probability that a customer purchases a product given that they clicked an advertisement.

That is a conditional probability problem.

Independent Events

Two events are independent if one does not affect the probability of the other.

For independent events:

$$P(A \cap B) = P(A) \times P(B)$$

Example:

- Tossing a coin
- Rolling a die

The result of the coin toss does not affect the die roll.

Probability Distributions

A probability distribution describes how probabilities are assigned to possible outcomes.

Important distributions include:

1. Bernoulli Distribution
2. Binomial Distribution
3. Poisson Distribution
4. Uniform Distribution
5. Normal Distribution

Bernoulli Distribution

Bernoulli distribution describes an experiment with **two possible outcomes**.

Examples:

- Success / Failure
- Yes / No
- 0 / 1
- Pass / Fail

Example:

Customer purchases $\rightarrow 1$
Customer does not purchase $\rightarrow 0$

Binomial Distribution

Binomial distribution describes the number of successes in a fixed number of independent Bernoulli trials.

Example:

Out of 100 customers, how many purchase a product?

Each customer can be treated as:

Purchase = Success
No Purchase = Failure

Poisson Distribution

Poisson distribution is commonly used to model the number of events occurring within a fixed interval of time or space when events occur at a certain average rate.

Examples:

- Number of calls received per hour
- Number of website errors per minute
- Number of customers arriving per hour

Uniform Distribution

In a uniform distribution, outcomes in a specified range have equal density.

For example, a continuous uniform distribution between 0 and 1 gives equal density across that interval.

Normal Distribution

The normal distribution is one of the most important distributions in statistics.

It is widely used because many measurements can be approximately normal under appropriate conditions.

Examples can include:

- Measurement errors
- Heights within a suitable population
- Test scores under certain conditions

2.What Is Inferential Statistics?

Inferential statistics uses **sample data to make conclusions or estimates about a larger population.**

It answers:

“What can I conclude about the population based on my sample?”

Example:

A company has 1 million customers.

Instead of surveying all 1 million customers, we select 2,000 customers.

We analyze those 2,000 customers and use statistical methods to make conclusions about the larger customer population.

Why Do We Need Inferential Statistics?

Because studying an entire population is often:

- Expensive

- Time-consuming
- Impractical
- Sometimes impossible

Inferential statistics allows us to make informed conclusions using samples.

Sampling

Sampling means selecting observations from a population.

Important sampling methods include:

Simple Random Sampling

Every member has a known and generally equal chance of selection.

Stratified Sampling

The population is divided into groups called strata, and samples are taken from each group.

Example:

Male
Female
Different age groups
Different regions

Systematic Sampling

Select every k th observation after a starting point.

Example:

Every 10th customer

Cluster Sampling

Divide the population into clusters and select some clusters.

Sampling Bias

Sampling bias occurs when the sample systematically differs from the population in a way that affects conclusions.

Example:

Suppose we want to know how satisfied all university students are.

If we survey only students from one department, our sample may not represent the entire university.

Potential sources of bias include:

- Selection bias
- Non-response bias
- Undercoverage
- Convenience sampling

A poor sample can produce misleading conclusions even if the statistical calculations are correct.

Sampling Distribution

A sampling distribution is the distribution of a statistic calculated from **many possible samples** of the same size from a population.

For example:

Imagine repeatedly taking samples of 100 customers and calculating the mean purchase amount each time.

The collection of those sample means forms a sampling distribution of the sample mean.

Central Limit Theorem (CLT)

The Central Limit Theorem is one of the most important concepts in inferential statistics.

Under appropriate conditions, as sample size becomes sufficiently large, the sampling distribution of the sample mean tends to become approximately normal, even when the original population is not normally distributed.

This is extremely useful because it allows us to use normal-based methods for many sampling problems.

The required sample size depends on the underlying population distribution and other conditions; there is no single universal cutoff that works for every problem.

Standard Error

Standard error measures the variability of a sample statistic across repeated samples.



For the sample mean:

$$SE = \sigma / \sqrt{n}$$

When population standard deviation is unknown, we often estimate it using:

$$SE \approx s / \sqrt{n}$$

As sample size increases, standard error generally decreases.

This means larger samples usually give more precise estimates of the population mean.

43. Standard Deviation vs Standard Error

Standard Deviation

Describes variability of observations

Focuses on the data

Usually does not automatically decrease just because n increases

Used in descriptive statistics

Standard Error

Describes variability of a sample statistic

Focuses on the estimate

Often decreases as sample size increases

Important in inferential statistics

Difference between Standard deviation and Standard error

Point Estimation

A point estimate is a single value used to estimate a population parameter.

Examples:

Sample mean $\rightarrow$ estimate of population mean

Sample proportion $\rightarrow$ estimate of population proportion

For example:

If the average income in our sample is ₹40,000, then ₹40,000 can be used as a point estimate of the population mean income.

Confidence Intervals

A confidence interval provides a range of plausible values for a population parameter based on sample data.

Example:

95% Confidence Interval = ₹38,000 to ₹42,000

A confidence interval communicates both:

- An estimate
- Uncertainty around that estimate

Correct Interpretation of a 95% Confidence Interval

A common mistake is saying:

“There is a 95% probability that the true population mean is inside this particular interval.”

That is not the standard frequentist interpretation.

A better interpretation is:

If we repeatedly took samples and constructed confidence intervals using the same procedure, approximately 95% of those intervals would contain the true population parameter, assuming the procedure's conditions are satisfied.

For a single computed interval, we typically say that the interval provides a **95% confidence level** for the parameter.

Margin of Error

Margin of error represents the amount by which an estimate may differ from the true population parameter under the chosen confidence procedure.

A simplified confidence interval can be expressed as:

Estimate $\pm$ Margin of Error

For example:

Mean = 50
Margin of Error = 3

Then:

Confidence Interval = 47 to 53

Hypothesis Testing

Hypothesis testing is a statistical framework for evaluating a claim about a population using sample data.

Example:

A company claims:

“Our new website increases average customer spending.”

We collect sample data and test whether the evidence supports a difference or increase.

Null and Alternative Hypotheses

Hypothesis testing usually begins with two hypotheses.

Null Hypothesis — H_0

Represents the baseline or no-effect claim.

Example:

$$H_0: \mu = 500$$

Alternative Hypothesis — H_1 or H_a

Represents the effect or difference we are testing for.

Example:

$$H_1: \mu \neq 500$$

Depending on the research question, the alternative could instead be:

$$H_1: \mu > 500$$

or

$$H_1: \mu < 500$$

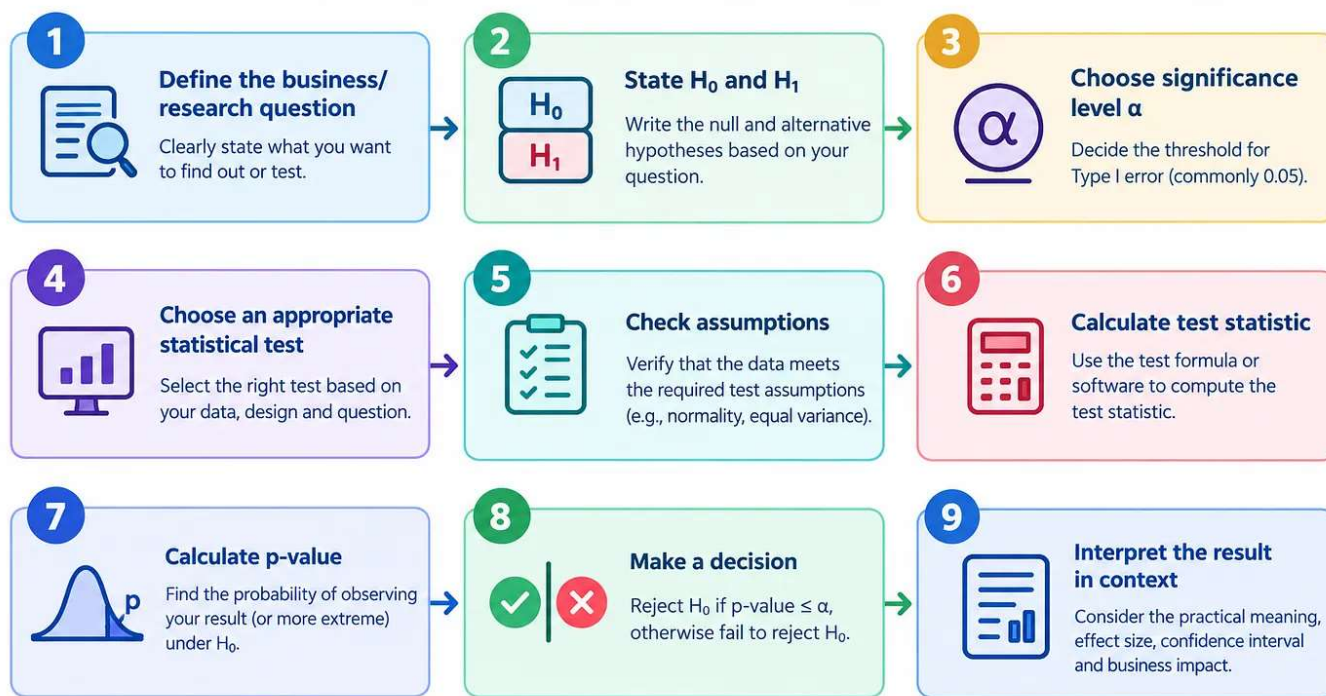
Hypothesis Testing Workflow

A typical workflow is:



Hypothesis Testing Workflow

From question to decision – step by step



Hypothesis Testing Workflow — From Defining the Research Question to Interpreting the Results

Significance Level (α)

The significance level is the threshold chosen before the test for deciding how much evidence is required against H_0 .

A common choice is:

$$\alpha = 0.05$$

This corresponds to a 5% Type I error rate in the long-run interpretation of the testing procedure, under its assumptions.

P-value

The p-value is one of the most misunderstood concepts in statistics.

A p-value is calculated **assuming the null hypothesis is true**.

It measures how compatible the observed data, or data more extreme than what was observed, are with the null hypothesis.

A small p-value means the observed result would be relatively unusual if H_0 were true.

Example

Suppose:

$$\begin{aligned} \text{p-value} &= 0.02 \\ \alpha &= 0.05 \end{aligned}$$

Since:

$$0.02 < 0.05$$

we reject H_0 .

Important

A p-value is **not**:

“The probability that H_0 is true.”

Reject vs Fail to Reject

If:

$$\text{p-value} < \alpha$$

we typically:

Reject H_0

If:

$$\text{p-value} \geq \alpha$$

we typically:

Fail to reject H_0

We should avoid saying:

“We proved H_0 is true.”

Failing to reject H_0 means the sample did not provide sufficient evidence against it at the chosen significance level.

One-Tailed vs Two-Tailed Tests

Two-Tailed Test

Used when we want to detect a difference in either direction.

$$H_1: \mu \neq \mu_0$$

Right-Tailed Test

Used when we are testing whether the parameter is greater.

$$H_1: \mu > \mu_0$$

Left-Tailed Test

Used when we are testing whether the parameter is smaller.

$$H_1: \mu < \mu_0$$

The choice should be based on the research question and decided appropriately rather than selected after seeing the results.

Type I Error

Type I error occurs when we:

Reject a true null hypothesis.

In simple terms: In simple terms:

False Positive

The significance level α controls the long-run Type I error rate under the assumptions of the test.

Type II Error

Type II error occurs when we:

Fail to reject a false null hypothesis

In simple terms:

False Negative

It is commonly represented by:

β

Statistical Power

Statistical power is the probability of correctly detecting an effect when a specified alternative is true.

It is commonly expressed as:

$$\text{Power} = 1 - \beta$$

Power generally increases with:

- Larger sample size
- Larger true effect
- Lower variability
- Appropriate significance level choices

Effect Size

Statistical significance tells us whether the evidence is strong enough under a specified testing framework.

Effect size tells us:

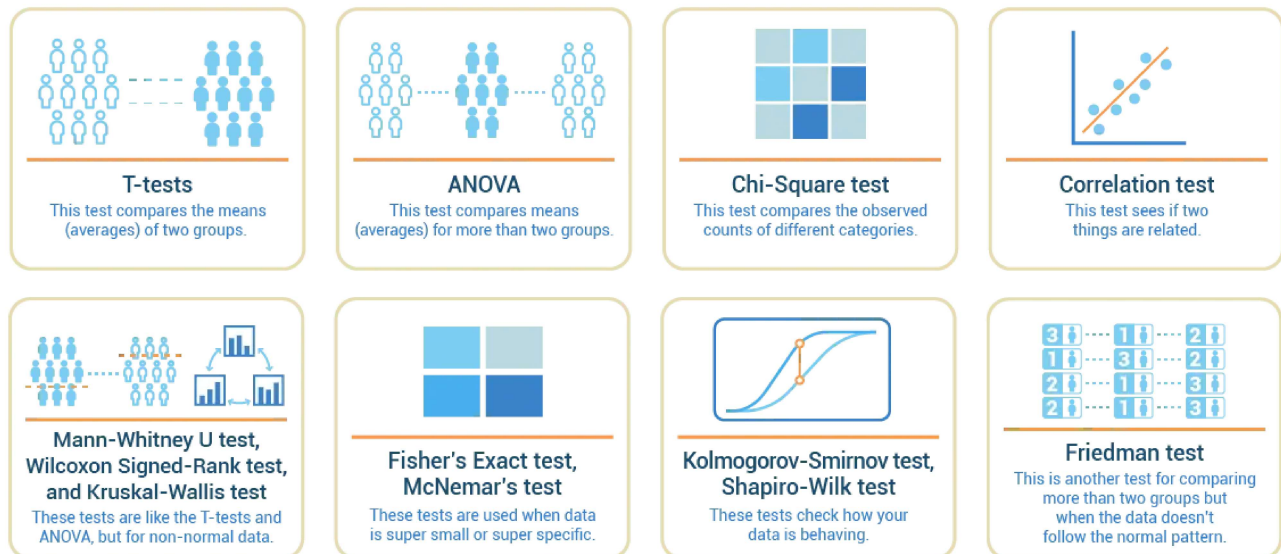
How large is the difference or relationship?

This is important because a very large sample can make a tiny difference statistically significant.

Examples of effect-size measures include:

- Cohen's d
- Correlation coefficient
- Odds ratio
- Risk ratio

Common Statistical Tests



Statistical Tests

Z-Test

A z-test is commonly used when:

- The population standard deviation is known, or

- Conditions make a normal-based test appropriate

For a one-sample mean:

$$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$$

In practice, when σ is unknown and estimated from the sample, a t-test is often more appropriate.

T-Test

The t-test is commonly used for comparing means when population standard deviation is unknown.

Common types:

1. One-sample t-test
2. Independent two-sample t-test
3. Paired t-test

One-Sample T-Test in Python

```
from scipy.stats import ttest_1samp
```

```
sample = [510, 520, 495, 505, 515, 500, 525]
```

```
statistic, p_value = ttest_1samp(sample, 500)
```

```
print("Test Statistic:", statistic)
print("P-value:", p_value)
```

Interpret the p-value using the significance level selected for your analysis.

Independent T-Test in Python

Suppose we want to compare the average salaries of two independent groups.

```
from scipy.stats import ttest_ind
```

```
group_a = [40, 42, 45, 43, 44]
group_b = [48, 50, 47, 49, 51]

statistic, p_value = ttest_ind(
    group_a,
    group_b,
    equal_var=False
)

print("Test Statistic:", statistic)
print("P-value:", p_value)
```

Using:

```
equal_var=False
```

performs Welch's t-test, which does not assume equal population variances.

Paired T-Test

A paired t-test is used when observations are naturally paired.

Example:

- Before training score
- After training score

for the same employees.

```
from scipy.stats import ttest_rel

before = [60, 65, 70, 72, 68]
after = [68, 70, 75, 78, 74]

statistic, p_value = ttest_rel(before, after)

print("Test Statistic:", statistic)
print("P-value:", p_value)
```

Chi-Square Test

The chi-square test is commonly used with categorical data.

Example question:

Is customer preference associated with gender?

```
from scipy.stats import chi2_contingency
```

```
table = [  
    [50, 30],  
    [40, 45]  
]  
  
chi2, p_value, dof, expected = chi2_contingency(table)  
  
print("Chi-square:", chi2)  
print("P-value:", p_value)
```

ANOVA

ANOVA stands for:

Analysis of Variance

It is commonly used to compare the means of three or more groups.

Example:

```
Group A  
Group B  
Group C
```

The null hypothesis is that the group means are equal.

```
from scipy.stats import f_oneway  
  
group_a = [10, 12, 11, 13, 12]  
group_b = [15, 14, 16, 15, 17]  
group_c = [20, 21, 19, 22, 20]  
  
statistic, p_value = f_oneway(  

```

```
group_a,  
group_b,  
group_c  
)  
  
print("F-statistic:", statistic)  
print("P-value:", p_value)
```

Post-Hoc Tests

If ANOVA indicates that there is evidence of a difference somewhere among multiple groups, ANOVA alone does not tell us exactly which groups differ.

A post-hoc test can help identify the pairwise differences.

One common method is **Tukey's HSD**.

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd  
  
values = group_a + group_b + group_c  
  
labels = (  
    ["A"] * len(group_a) +  
    ["B"] * len(group_b) +  
    ["C"] * len(group_c)  
)  
  
result = pairwise_tukeyhsd(  
    values,  
    labels  
)  
  
print(result)
```

Non-Parametric Tests

Non-parametric tests are useful in situations where the assumptions of common parametric tests are not appropriate.

Examples include:

- Mann–Whitney U test
- Wilcoxon signed-rank test
- Kruskal–Wallis test

However, non-parametric does **not** mean “assumption-free.”

These tests have their own assumptions and conditions.

Mann–Whitney U Test

The Mann–Whitney U test is commonly used for comparing two independent groups when a rank-based approach is appropriate.

```
from scipy.stats import mannwhitneyu

group_a = [10, 12, 15, 13, 14]
group_b = [20, 22, 18, 21, 19]

statistic, p_value = mannwhitneyu(
    group_a,
    group_b,
    alternative="two-sided"
)

print("U Statistic:", statistic)
print("P-value:", p_value)
```

Wilcoxon Signed-Rank Test

The Wilcoxon signed-rank test is commonly used for paired data when a non-parametric alternative to the paired t-test is appropriate.

```
from scipy.stats import wilcoxon

before = [60, 65, 70, 72, 68]
after = [68, 70, 75, 78, 74]

statistic, p_value = wilcoxon(before, after)

print("Statistic:", statistic)
print("P-value:", p_value)
```

Kruskal-Wallis Test

The Kruskal-Wallis test is commonly used to compare three or more independent groups using a rank-based approach.

```
from scipy.stats import kruskal

group_a = [10, 12, 11, 13, 12]
group_b = [15, 14, 16, 15, 17]
group_c = [20, 21, 19, 22, 20]

statistic, p_value = kruskal(
    group_a,
    group_b,
    group_c
)

print("Statistic:", statistic)
print("P-value:", p_value)
```

If the result is significant, post-hoc analysis may be needed to identify which groups differ.

Covariance

Covariance measures the direction in which two variables change together.

If:

$$\text{Covariance} > 0$$

the variables tend to increase together.

If:

$$\text{Covariance} < 0$$

one tends to increase when the other decreases.

If covariance is near zero, there may be little linear co-movement.

However, covariance depends on the units of measurement, making it harder to compare across datasets.

Correlation

Correlation measures the **strength and direction of a linear relationship** between two variables.

The correlation coefficient usually ranges from:

-1 to +1

Interpretation:

+1 → Perfect positive linear relationship
0 → No linear correlation
-1 → Perfect negative linear relationship

Example:

Study Hours ↑
Exam Score ↑

This may result in positive correlation.

Correlation Does Not Mean Causation

This is one of the most important concepts in statistics.

Suppose:

Ice cream sales ↑
Swimming incidents ↑

Both may increase together.

Does ice cream cause swimming incidents?

Not necessarily.

A third factor such as:

Hot weather

may influence both.

Therefore:

Correlation does not prove causation.

Regression

Regression is used to model relationships between variables and make predictions or estimate effects under appropriate assumptions.

A simple linear regression equation is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Where:

- y = dependent variable
- x = independent variable
- β_0 = intercept
- β_1 = slope
- ε = error term

Example:

$$\text{House Price} = \beta_0 + \beta_1 \times \text{Area} + \varepsilon$$

The coefficient β_1 represents the expected change in the outcome for a one-unit increase in x , conditional on the model and its assumptions.

Statistical Test Assumptions

Before applying a statistical test, we should understand its assumptions.

Depending on the test, assumptions can include:

- Independence

- Appropriate measurement scale
- Approximate normality
- Equal variance
- Random or appropriate sampling
- Appropriate study design

For example, a standard independent t-test traditionally assumes equal variances, while Welch's t-test relaxes that assumption.

Checking assumptions helps prevent misleading conclusions.

Multiple Testing Problem

Suppose we perform one statistical test at:

$$\alpha = 0.05$$

The chance of a false positive is controlled at 5% under the testing framework.

But if we perform many tests, the probability of getting at least one false positive can increase.

This is called the:

Multiple Testing Problem

Common correction approaches include:

Bonferroni Correction

A simple adjustment that controls the family-wise error rate.

Holm Correction

A step-down method that is often less conservative than simple Bonferroni.

False Discovery Rate (FDR)

Methods such as Benjamini–Hochberg control the expected proportion of false discoveries among rejected hypotheses under specified conditions.

A/B Testing

A/B testing is a practical application of statistical inference.

Suppose an e-commerce company has two versions of a webpage.

Version A → Existing page
Version B → New page

Users are randomly assigned to the two versions.

We compare a metric such as:

Conversion Rate

Example:

A → 5.2%

B → 5.8%

We need to determine whether the observed difference is likely to reflect a real difference rather than random sampling variation.

A/B testing can involve:

- Hypothesis testing
- Confidence intervals
- Effect sizes
- Sample size planning
- Statistical power

Statistical Significance vs Business Significance

These are not the same.

Suppose a new feature increases conversion rate from:

10.00% → 10.05%

With a huge sample, the difference might be statistically significant.

But the business impact may be too small to justify the engineering or operational cost.

Confidence Interval vs Prediction Interval

A confidence interval is typically used to estimate a **population parameter**.

A prediction interval is used to estimate where a **future individual observation** may fall.

Example:

Confidence Interval

Estimate the average salary of the population.

Prediction Interval

Predict the salary of one future individual based on a regression model.

Prediction intervals are generally wider because they include both uncertainty in the estimated mean response and individual-level variation.

Complete Statistical Example

Let's consider a real-world example.

Suppose a company has:

Population = 100,000 customers

We randomly select:

Sample = 1,000 customers

Suppose the average purchase amount in the sample is:

₹2,500

and the sample standard deviation is:

₹500

Step 1: Descriptive Statistics

We calculate:

Mean = ₹2,500

SD = ₹500

This describes our sample.

Step 2: Distribution

We visualize the purchase amounts.

We check:

- Shape
- Skewness
- Outliers
- Center
- Spread

Step 3: Sampling

Our sample of 1,000 customers is intended to represent the larger population.

The quality of the inference depends heavily on how the sample was obtained.

Step 4: Point Estimate

Our point estimate of the population mean is:

₹2,500

Step 5: Standard Error

Using:

$$SE \approx s / \sqrt{n}$$

we get:

$$SE \approx 500 / \sqrt{1000}$$

Approximately:

$$SE \approx ₹15.81$$

Step 6: Confidence Interval

For a 95% confidence interval, a large-sample normal approximation would use approximately:

$$\text{Estimate} \pm 1.96 \times SE$$

So:

$$₹2,500 \pm 1.96 \times ₹15.81$$

The approximate interval is:

$$₹2,469 \text{ to } ₹2,531$$

The exact method may differ depending on assumptions and the chosen statistical procedure.

Hypothesis Test in the Same Example

Suppose the company wants to test:

Is the true average purchase amount different from ₹2,400?

We define:

$$H_0: \mu = ₹2,400$$

$$H_1: \mu \neq ₹2,400$$

Our sample mean is:

₹2,500

We choose:

$\alpha = 0.05$

Then we select an appropriate test, calculate the test statistic and p-value, check assumptions, and interpret the result.

But we should not stop at the p-value.

We should also consider:

- Effect size
- Confidence interval
- Practical importance
- Business impact

Python Statistics Ecosystem

Python provides several powerful libraries for statistics.

NumPy

Used for numerical computing.

```
import numpy as np
```

Pandas

Used for data manipulation and descriptive analysis.

```
import pandas as pd
```

SciPy

Provides many statistical tests and scientific computing functions.

Statsmodels

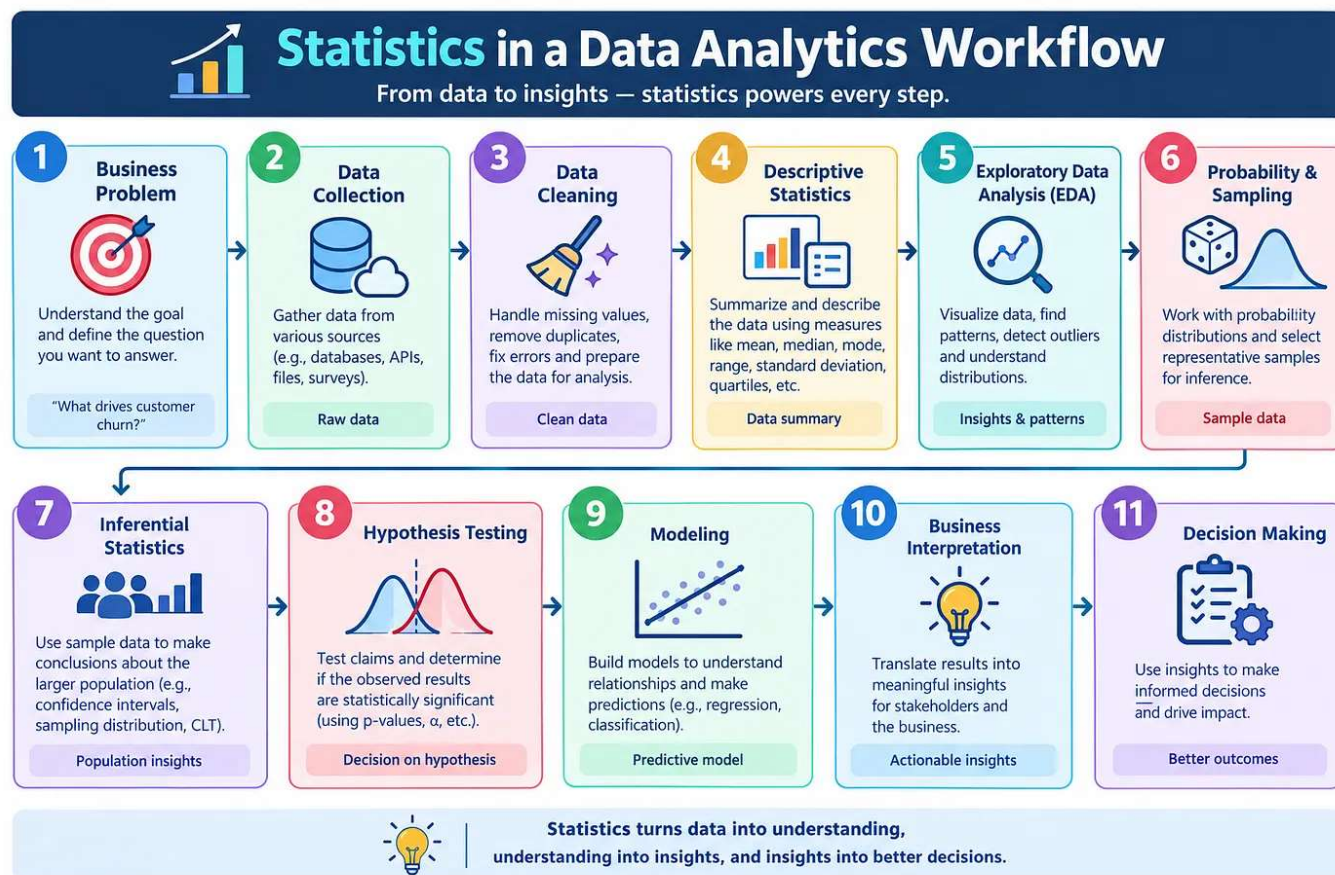
Useful for statistical models, regression, hypothesis testing, and detailed statistical output.

```
import statsmodels.api as sm
```

Scikit-learn

Primarily used for machine learning, but also provides tools for preprocessing, modeling, evaluation, and related statistical workflows.

```
import sklearn
```



Final Takeaway

Statistics is one of the most important foundations of Data Science and Data Analytics.

Descriptive statistics helps us understand what has happened in our data.

Inferential statistics helps us use sample data to make estimates and evaluate claims about a larger population.

The most important lesson is that statistics is not just about formulas.

It is about **understanding uncertainty and making better decisions from data.**



Written by Marishetty Ramyakrishna

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